

ESE 5700: Digital Integrated Circuits and VLSI Fundamentals

Project 1, Part 1: The 6T Bitcell

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Team and contributions

Member	Contributions (this part)
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Both team members are responsible for understanding the full design and report.

1. Sizing (β and γ)

Hand analysis

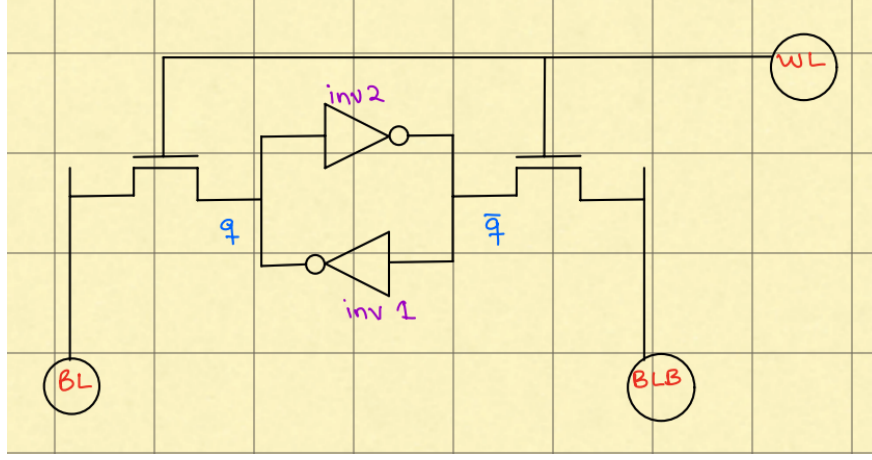


Figure 1: 6T cell read-disturb setup: $Q = 0$ stored, wordline asserted, bitlines precharged (hand sketch).

Read stability

Assume the stored state $Q = 0$ ($\bar{Q} \approx V_{DD}$) for read-stability analysis. With $Q = 0$ the “0” node is pulled down by the NMOS inside INV1 while the bitline (precharged to V_{DD}) drives current through the pass-gate, so the disturbed node follows an approximate voltage-divider model:

$$\frac{V_Q}{V_{DD}} = \frac{R_{PD1}}{R_{PG} + R_{PD1}} = \frac{1}{1 + R_{PG}/R_{PD1}}. \quad (1)$$

From transistor physics $R_T \propto 1/W_T$ at fixed L . Defining the cell ratio $\beta = (W/L)_{PD}/(W/L)_{PG} = W_{PD}/W_{PG}$ gives $\beta \approx R_{PG}/R_{PD}$, hence

$$\frac{V_Q}{V_{DD}} \approx \frac{1}{1 + \beta}, \quad V_Q \approx \frac{V_{DD}}{1 + \beta} : \quad (2)$$

a larger β lowers the read-disturb bump. Whether that bump flips the cell is set by the inverter switching voltage V_M ($V_{in} = V_{out}$), where both devices conduct.

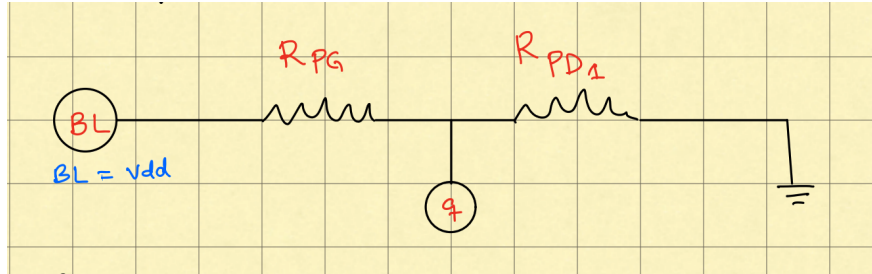


Figure 2: Approximate resistive divider for the read disturb: precharged bitline through R_{PG} against the pull-down R_{PD1} to ground (hand sketch).

Current balance at the equilibrium disturb voltage: neglecting gate leakage, KCL gives $I_{PD} = I_{PG}$. The pass-gate has $D = G = V_{DD}$, $S = V_Q$, so $V_{DS} = V_{GS} = V_{DD} - V_Q$ and $V_{DS} > V_{GS} - V_{Tn}$ always: saturation, with $I_{PG} = \frac{1}{2}k_{PG}(V_{GS} - V_{Tn})^2$ and $k_{PG} = \mu_n C_{ox}(W/L)_{PG}$. The pull-down (pd1) has $V_D = V_Q$, $V_G \approx V_{DD}$, $V_S = 0$, so $V_{DS} = V_Q < V_{GS} - V_{Tn}$: triode, with $I_{PD} = k_{PD}[(V_{GS} - V_{Tn})V_{DS} - V_{DS}^2/2]$. Equating the two with $E = V_{DD} - V_{Tn}$ yields

$$V_Q^2(1 + \beta) + V_Q(-2E - 2E\beta) + E^2 = 0 \quad \Rightarrow \quad V_Q = E \left(1 \pm \sqrt{\frac{\beta}{1 + \beta}} \right), \quad (3)$$

of which the positive root exceeds $V_{DD} - V_{Tn}$ and is inconsistent with the assumed saturated pass-gate / triode pull-down, leaving

$$V_Q = (V_{DD} - V_{Tn}) \left(1 - \sqrt{\frac{\beta}{1 + \beta}} \right). \quad (4)$$

This assumes ideal long-channel square-law behavior, equal NMOS thresholds, and negligible body effect.

The disturbed node Q drives INV2, whose switching point satisfies $I_{DN} = |I_{DP}|$ with both devices in saturation:

$$V_M = \frac{V_{Tn} + B(V_{DD} - |V_{Tp}|)}{1 + B}, \quad B = \sqrt{\frac{k_P}{k_N}} = \sqrt{\frac{\gamma \mu_p}{\beta \mu_n}}, \quad (5)$$

with $k_P/k_N = \mu_p C_{ox}(W/L)_{PU}/(\mu_n C_{ox}(W/L)_{PG})$, $\beta = (W/L)_{PD}$, and $\gamma = (W/L)_{PG}$ as written in the hand-note derivation, so V_M depends on the supply, the threshold voltages, and the relative PMOS/NMOS strengths. Read stability requires $V_Q < V_M$:

$$(V_{DD} - V_{Tn}) \left(1 - \sqrt{\frac{\beta}{1 + \beta}} \right) < \frac{V_{Tn} + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}} (V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}}}. \quad (6)$$

Increasing β strengthens the pull-down relative to the pass-gate and lowers the disturb voltage, but β also moves V_M and costs area: the final cell ratio is not selected by minimizing V_Q alone.

Write stability

Writability/ γ estimate: *TBD*.

Chosen sizes

Device	(W/L)	Notes
Pull-down (PD)	<i>TBD</i>	
Pull-up (PU)	<i>TBD</i>	
Pass-gate (PG)	<i>TBD</i>	
β	<i>TBD</i>	
γ	<i>TBD</i>	

β sweep (schematic)

β	SNM _{read}	Notes
1×	TBD	
1.5×	TBD	
2×	TBD	

Design point selected: *TBD*. Cell kept symmetric.

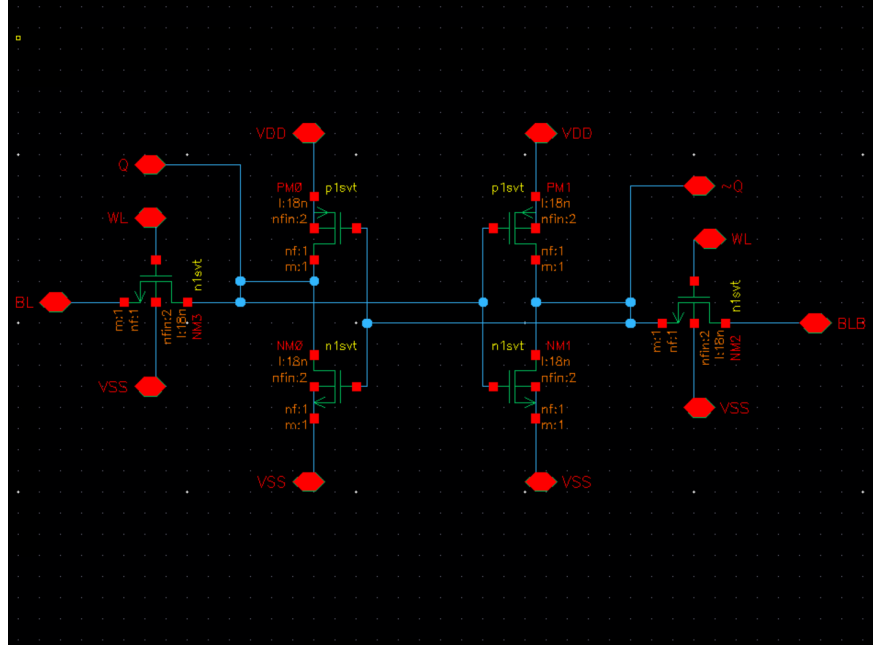


Figure 3: 6T bitcell schematic with transistor sizes (proj1_lib 6t-sram, Virtuoso capture).

2. Layout

DRC- and LVS-clean layout: wordline horizontal, bitline pair vertical, tileable for Part 2. Report area in μm^2 and F^2 ($F = 45\text{ nm}$).

Metric	Value
Cell area	<i>TBD</i> μm^2
Cell area in F^2	<i>TBD</i> F^2
DRC	<i>clean / summary</i>
LVS	<i>clean / summary</i>

3. Extraction

Post-layout parasitic extraction. Report per-cell bitline-node and wordline-node capacitances (loads for Part 2).

Capacitance	Value
C_{BL} (per cell)	<i>TBD</i>
C_{WL} (per cell)	<i>TBD</i>

4. Stability characterization (post-layout, $V_{DD} = 1.2\text{ V}$)

4a. Hold SNM

Butterfly plot with $WL = 0$; report $SNM_{\text{hold}} = \min(SNM_L, SNM_R)$.

4b. Read SNM

Butterfly plot with $WL = V_{DD}$, $BL=BLB= V_{DD}$; report SNM_{read} and DC voltage of the “0” node vs. the Section 1 estimate.

4c. Write margin

Store $Q = 1$, $WL = V_{DD}$, $BL = V_{DD}$; sweep BLB ; report $WM_{BL} = V_{\text{trip}}$.

4d. Cell current

Report I_{cell} from the “0”-side bitline ($WL = V_{DD}$, bitlines at V_{DD}).

4e. Supply dependence

V_{DD}	SNM_{read}	WM_{BL}
1.2 V	<i>TBD</i>	<i>TBD</i>
0.9 V	<i>TBD</i>	<i>TBD</i>
0.6 V	<i>TBD</i>	<i>TBD</i>

Still reads and writes at 0.6 V? *yes/no + notes.*

5. Functional check

Transient test with ideal WL and precharge pulses; bitline pair loaded by 15 additional per-cell bitline loads. Write 0, read; write 1, read. Show WL, BL, BLB, Q, $\bar{Q}$.

6. Discussion

Short paragraph: what limits the cell, and what you would change for Part 2.

Technology note

All work is done in the Cadence GPDK 18 nm technology (schematic, layout, Spectre/ADE, DRC/LVS/PEX) at nominal $V_{DD} = 1.2\text{ V}$ unless noted. Characterization above is post-layout unless a subsection states otherwise.